Lecture 11/12

Chapter 6: Continuous time Markov Chains

References: Oliver Ibe – Markov Processes for Stochastic Modeling (Chapter 5)

Sheldon Ross – Introduction to Probability Models (Chapter 6)

$$P[X(t+s) = j | X(s) = i, X(u) = x(u), 0 \leq u < s] = P[X(t+s) = j | X(s) = i]$$

Therefore, the quantities of interest associated with a continuous-time Markov chain are:

i) The distribution of the time interval that the process stays at the ith state before making a tansition to other state (not itself). In other words, the rate at which process makes transition from the ith state.

ii) Whenever a transition occurs from state i, after staying a time t in the state, the probability that the process will move on to state j. Let us denote the transition probability from state i to state j as p_{ij}.

Let T_i denote the probability that the process stays at state i before making a transition to a different state. Then, at time t, we have two different scenarios to consider.

i) We keep track of the time interval t that the process already stayed at state i, and calculate the total time the process stays at state i. Then, the probability that the process makes a transition from state i, given that it already stayed there for a period of time t, is $P[T_i > t + s | T_i > t]$.

ii) We forget the time the process already stayed t period in state i, and only consider the reamining time after which it will make a transition. More specifically, if the process currently in state i, then we always consdier the process has just entered into state i, and the interval the process stays in state i starts from now. In other words, consider a new process which enters into state i at time t, and we are interested in to know the probability $P[T_i > s]$ that the process will make a transition after a period s.

Now, according to the Markovian property for a continuous-time Markov chain, the conditional distribution of the transition probabilities only depends on the present, which includes

a) the present state of the process, and

b) the interval of time starting from the present time for which we want to calculate the probability.

$$P[T_i > t + s | T_i > t] = P[T_i > s],$$

The above observations lead us to an alternate definition of the continuous-time Markov chain. A continuous-time Markov chain is a stochastic process which, after each transition to state i, follows the following properties:

i) the amount of time it stays in state i before making a transition to a different state is exponentially distributed with parameter v_i, i.e., the expected amount of time it stays in state i is $1/v_i$.

ii) when the process leaves state i, it next enters state j with probability p_{ij}, which satisfies

$$\begin{aligned} p_{ii} &= 0, && \forall i \geq 0 \\ \sum_j p_{ij} &= 1, && \forall i \geq 0 \end{aligned}$$

iii) the amount of time the process spends in the current state and the state the process will visit next are independent random variables, otherwise, we need to keep track of time that the process is staying in the current state to calculate the probability p_{ij} which, as we already proved, contradicts with the Markovian property.

Ex 4.1.2 Consider a shoeshine shop consisting of 2 chairs–chair 1 and chair 2. A customer upon arrival goes initially to chair 1 and then to chair 2. The service times at two chairs are assumed to be independent rvs which are exponentially distributed with service rates μ_1 and μ_2. Suppose that potential customers arrive in accordance with a Poisson process having rate λ, and that a potential customers will only enter the system if both chairs are empty. Model the system state by CTMC.

Sol: $v_0 = \lambda, v_1 = \mu_1, v_2 = \mu_2$. $P_{01} = P_{12} = P_{20} = 1$

Transition Probability Function

$$p_{ij}(t) = P[X(t+s) = j | X(s) = i]$$

$$p_j(t) = P[X(t) = j]$$

$$\sum_j p_{ij}(t) = 1$$

$$\sum_j p_j(t) = 1$$

Chapman-Kolmogorov Equations

$$p_{ij}(s+t) = \sum_k p_{ik}(s) p_{kj}(t)$$

$$
\begin{aligned}
P_{ij}(t+s) &= P[X(t+s)=j \mid X(0)=i] \\
&= \sum_{k=0}^{\infty} P[X(t+s)=j, X(t)=k \mid X(0)=i] \\
&= \sum_{k=0}^{\infty} P[X(t+s)=j \mid X(t)=k, X(0)=i] P[X(t)=k \mid X(0)=i] \\
&= \sum_{k=0}^{\infty} P[X(t+s)=j \mid X(t)=k] P[X(t)=k \mid X(0)=i] \\
&= \sum_{k=0}^{\infty} P_{kj}(s) P_{ik}(t)
\end{aligned}
$$

$$P(t) = \begin{bmatrix} p_{11}(t) & p_{12}(t) & p_{13}(t) & \dots \\ p_{21}(t) & p_{22}(t) & p_{23}(t) & \dots \\ p_{31}(t) & p_{32}(t) & p_{33}(t) & \dots \\ \dots & \dots & \dots & \dots \end{bmatrix}$$

$$P(t+s) = P(t)P(s)$$

Dwell Time/Holding Time

Time that a MC stayed at state i before making a transition

T_i => dwell time

$T_i \sim Exp\ (v_i)$

$E[T_i] = \frac{1}{v_i}$

After the dwell time a transition occurs to sate j, with probability p_{ij}

$\sum_j p_{ij} = 1$

State Transition Rate

v_i => rate at which process leaves state i

$v_i p_{ij}$ => rate at which makes a transition to state j

Probability of a transition in a small time Δt, from i to j

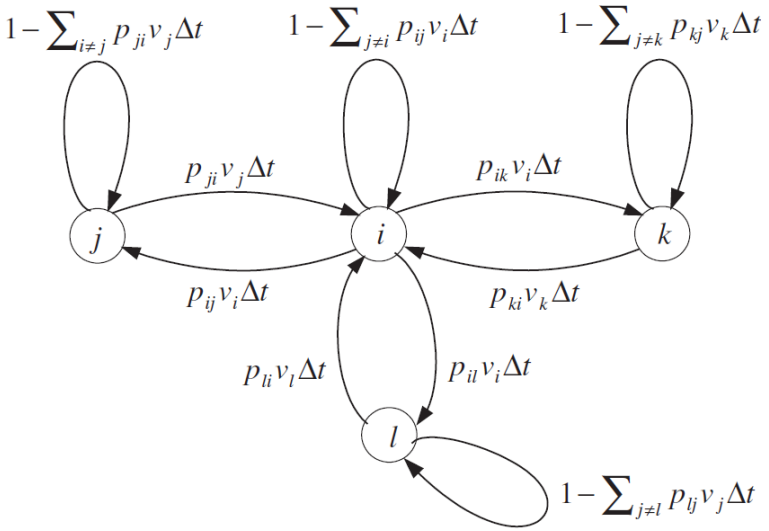
$$v_i p_{ij} \Delta t$$

Prob. of a transition from i to any state j

$$\sum_{j \neq i} v_i p_{ij} \Delta t$$

Prob. of no Transition from i

$$1 - \sum_{j \neq i} v_i p_{ij} \Delta t$$



$$p_i(t+\Delta t) = p_i(t)\left\{1 - \sum_{j\neq i} p_{ij}v_i\Delta t\right\} + \sum_{j\neq i} p_j(t)p_{ji}v_j\Delta t$$

$$p_i(t+\Delta t) - p_i(t) = -\,p_i(t)\sum_{j\neq i} p_{ij} v_i \Delta t + \sum_{j\neq i} p_j(t) p_{ji} v_j \Delta t$$

$$\frac{p_i(t+\Delta t) - p_i(t)}{\Delta t} = -\,v_i p_i(t)\sum_{j\neq i} p_{ij} + \sum_{j\neq i} p_j(t) p_{ji} v_j$$

$$\lim_{\Delta t\to 0}\left\{\frac{p_i(t+\Delta t) - p_i(t)}{\Delta t}\right\} = \frac{\mathrm{d}p_i(t)}{\mathrm{d}t} = -\,v_i p_i(t)\sum_{j\neq i} p_{ij} + \sum_{j\neq i} p_j(t) p_{ji} v_j$$

In the steady state, $p_j(t) \rightarrow p_j$ and

$$\lim_{t \rightarrow \infty} \left\{ \frac{\mathrm{d}p_i(t)}{\mathrm{d}t} \right\} = 0$$

$$0 = -v_i p_i(t) \sum_{j \neq i} p_{ij} + \sum_{j \neq i} p_j(t) p_{ji} v_j$$

$$1 = \sum_i p_i$$

$$v_i p_i(t) \sum_{j \neq i} p_{ij} = \sum_{j \neq i} p_j(t) p_{ji} v_j$$

$$1 = \sum_i p_i$$

Birth and Death Processes

A birth and death process is a CTMC with states $\{0, 1, \cdots\}$ for which transitions from state n may go only to either state $n-1$ or state $n+1$.

Ex 4.2.1 Suppose there are n people in the system, then

i. new arrivals enter the system at an exponential rate λ_n, and

ii. people leave the system at an exponential rate μ_n.

This process is a birth and death process and the state transition rates and probabilities are

$$\begin{aligned} v_0 &= \lambda_0, \\ v_i &= \lambda_i + \mu_i, \quad i > 0 \\ P_{01} &= 1, \\ P_{i,i+1} &= \frac{\lambda_i}{\lambda_i + \mu_i}, \quad i > 0 \\ P_{i,i-1} &= \frac{\mu_i}{\lambda_i + \mu_i}, \quad i > 0 \end{aligned}$$

The parameters $\{\lambda_0, \lambda_1, \lambda_2, \cdots\}$ and $\{\mu_0, \mu_1, \mu_2, \cdots\}$ are called the arrival (or birth) and departure (or death) rates, respectively.

Ex 4.2.2 Consider a birth and death process for which

$$\mu_n = 0, \qquad \text{for all } n \geq 0$$

$$\lambda_n = \lambda, \qquad \text{for all } n \geq 0$$

This is a process in which departures never occur, and the time between successive arrivals is exponential with mean $1/\lambda$. Hence, this is just the Poisson process.

Ex 4.2.3 (M/M/1 Queueing System) Suppose that customers arrive at a single-server service station with a Poisson process having rate λ. Upon arrival, each customer goes directly into service if the server is free; if not, then the customer joins the queue. When the server finishes serving a customer, the customer leaves the system and the next customer in line, if there are any waiting, enters the service. This is known as M/M/1 queueing system.

If we let $X(t)$ denote the number of customers in the system at time t then $\{X(t), t \geq 0\}$ is a birth and death process with

$$\mu_n = \mu, \qquad n \geq 1$$
$$\lambda_n = \lambda, \qquad n \geq 0$$

Balance Equation for Birth and Death process

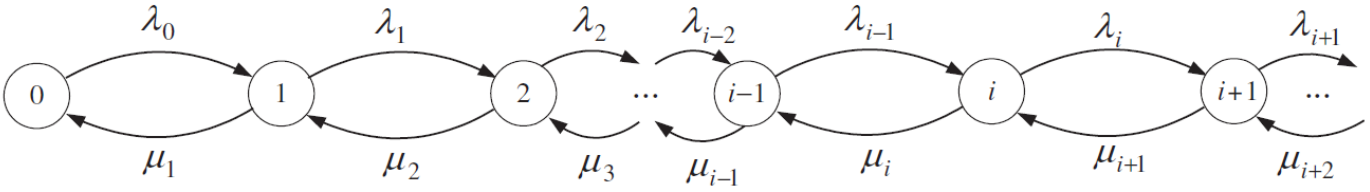


Figure 5.3 State-transition-rate diagram for birth and death process.

$$\lambda_i = v_i p_{i(i+1)}$$

$$\mu_i = v_i p_{i(i-1)}$$

$$\lambda_i + \mu_i = v_i$$

Transition from state i

$$p_{i(i+1)} = \lambda_i / (\lambda_i + \mu_i)$$

$$p_{i(i-1)} = \mu_i / (\lambda_i + \mu_i)$$

$$\frac{\mathrm{d}p_i(t)}{\mathrm{d}t} = -v_i p_i(t) \sum_{j \neq i} p_{ij} + \sum_{j \neq i} p_j(t) p_{ji} v_j$$

$$= -(\lambda_i + \mu_i) p_i(t) + \mu_{i+1} p_{i+1}(t) + \lambda_{i-1} p_{i-1}(t)$$

Thus, for the birth and death process we have that

$$\frac{\mathrm{d}p_0(t)}{\mathrm{d}t} = -\lambda_0 p_0(t) + \mu_1 p_1(t)$$

$$\frac{\mathrm{d}p_i(t)}{\mathrm{d}t} = -(\lambda_i + \mu_i)p_i(t) + \mu_{i+1}p_{i+1}(t) + \lambda_{i-1}p_{i-1}(t) \quad i > 0$$

In the steady state,

$$\lim_{t \to \infty} \left\{ \frac{\mathrm{d}p_i(t)}{\mathrm{d}t} \right\} = 0$$

$$\begin{aligned} \lambda_0 p_0 &= \mu_1 p_1 \\ (\lambda_i + \mu_i) p_i &= \mu_{i+1} p_{i+1} + \lambda_{i-1} p_{i-1} \qquad i = 1, 2, \ldots \\ \sum_i p_i &= 1 \end{aligned}$$

Balance Equation

Local Balance Equation

$$\lambda_0 p_0 = \mu_1 p_1$$
$$(\lambda_i + \mu_i)p_i = \mu_{i+1} p_{i+1} + \lambda_{i-1} p_{i-1} \quad i = 1, 2, \ldots$$
$$\sum_i p_i = 1$$

For $i = 1$, we obtain $(\lambda_1 + \mu_1)p_1 = \mu_2 p_2 + \lambda_0 p_0$. Because we know from the first equation that $\lambda_0 p_0 = \mu_1 p_1$, this equation becomes

$$\lambda_1 p_1 = \mu_2 p_2$$

Similarly, for $i = 2$, we have that $(\lambda_2 + \mu_2)p_2 = \mu_3 p_3 + \lambda_1 p_1$ Applying the last result we obtain

$$\lambda_2 p_2 = \mu_3 p_3$$

Repeated application of this method yields the general result

$$\lambda_i p_i = \mu_{i+1} p_{i+1} \qquad i = 0, 1, 2, \ldots$$

For any pair of states i and j, let

$$q_{ij} = v_i P_{ij}$$

$$v_i = \sum_j v_i P_{ij} = \sum_j q_{ij}$$

$$P_{ij} = \frac{q_{ij}}{v_i} = \frac{q_{ij}}{\sum_j q_{ij}}$$

Limiting Probabilities

In analogy with DTMC, the probability that a CTMC will be in state j at time t often converges to a limiting value which is independent of the initial state:

$$P_i = \lim_{t \to \infty} P_i(t)$$

Solution of the system of equations

$$v_j P_j = \sum_{k \neq j} q_{kj} P_k, \qquad \text{all states } j$$

$$\sum_j P_j = 1$$

A sufficient condition for existence of limiting probabilities P_j is that

i. All states of the MC communicate.

ii. The MC is positive recurrent.

$$v_j P_j = \text{rate at which the process leaves state } j$$

$$\sum_{k \neq j} q_{kj} P_k = \text{rate at which the process enters state } j$$

just a statement of the equality of the rates at which the process enters and leaves the state j. Because it balances these rates, the Eq. (4.12) are sometimes referred to as "balance equations."

State	Rate at which leave = rate at which enter
0	$\lambda_0 P_0 = \mu_1 P_1$
1	$(\lambda_1 + \mu_1) P_1 = \mu_2 P_2 + \lambda_0 P_0$
2	$(\lambda_2 + \mu_2) P_2 = \mu_3 P_3 + \lambda_1 P_1$
$\cdots$	$\cdots$
$n, n \geq 1$	$(\lambda_n + \mu_n) P_n = \mu_{n+1} P_{n+1} + \lambda_{n-1} P_{n-1}$

By adding to each equation the equation preceding it, we obtain

$$\begin{aligned}
\lambda_0 P_0 &= \mu_1 P_1, \\
\lambda_1 P_1 &= \mu_2 P_2, \\
\lambda_2 P_2 &= \mu_3 P_3, \\
&\cdots \\
\lambda_n P_n &= \mu_{n+1} P_{n+1}, \quad n \geq 0
\end{aligned}$$

$$
\begin{aligned}
P_1 &= \frac{\lambda_0}{\mu_1} P_0, \\
P_2 &= \frac{\lambda_1}{\mu_2} P_1 = \frac{\lambda_1 \lambda_0}{\mu_2 \mu_1} P_0, \\
P_3 &= \frac{\lambda_2}{\mu_3} P_2 = \frac{\lambda_2 \lambda_1 \lambda_0}{\mu_3 \mu_2 \mu_1} P_0, \\
&\cdots \\
P_n &= \frac{\lambda_{n-1}}{\mu_n} P_{n-1} = \frac{\lambda_{n-1} \lambda_{n-2} \cdots \lambda_1 \lambda_0}{\mu_n \mu_{n-1} \cdots \mu_2 \mu_1} P_0
\end{aligned}
$$

And by using the fact that $\sum_{n=0}^{\infty} P_n = 1$, we obtain

$$1 = P_0 + P_0 \sum_{n=1}^{\infty} \frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1}$$

$$P_0 = \frac{1}{1 + \sum_{n=1}^{\infty} \frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1}}$$

$$P_n = \frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1 \left(1 + \sum_{n=1}^{\infty} \frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1}\right)}, \quad n \geq 1$$

From the above, the necessary condition for the limiting probabilities to exist is

$$\sum_{n=1}^{\infty} \frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1} < \infty$$

This condition also may be shown to be sufficient.

Ex 4.4.2 (Ex 6.15, pp. 374) Consider the showshine shop and determine the proportion of the time the process is in each of the states 0, 1, 2. Note that it is not a birth and death process.